Blessed G/Michael Catholic Secondary School
1st Semester Chemistry mid Two for Grade 12th

December, 2018 E.C **Time Allowed :- 1:00hr**

Name Hanna Ayalew NO 24 Section 12th C

24/25 Excellent

R= 8.314J/ mol K = 0.0821Latm/ mol K

1F= 96500C/ mol e

Molar masses : Al = 27 g/ mol , Ag = 108 g/mol , O_2 = 32 g/mol H_2 =2g/mol

I. TRUE OR FALSE ITEM

Write true if the statement is correct and false if it is incorrect

False 1. The electronic conductivity increases with the increase in temperature

True 2. Electrolysis is used to bring chemical change in electrolytic cells and this process is not spontaneous.

True 3. In a galvanic cell, the anode mass must always decrease as the cell operates

True 4. The SHE can act as either anode or cathode depending on the cell it is paired with

False 5. The nature of the electrode has no effect on preferential discharge

II. MATCHING

Column A	Column B
A 6. SHE	A. Electrode with assigned potential of 0 volts
E 7. Standard electrode potential	B. species that cause oxidation by being reduced
B 8. Oxidizing agent	C. species that cause reduction by being oxidized
C 9. Reducing agent	D. difference between cathode and anode potentials
D 10. Cell potential (E^{o}_{cell})	E. Tendency of a half cell to be reduced at standard Conditions

III. MULTIPLE CHOICE QUESTION

Choose the best answer from the given alternatives.

C 11. The desired amount of charge for obtaining one mole of Al from Al^{+3} is

A. 96500 C B. 96500/3 C C. 3 * 96500 C D. 96500/2 C

A 12. In electrolysis of aqueous ~~copper sulphate~~ Lil. Na on solution using graphite electrode, the gas at anode and cathode respectively are

A. O_2 and H_2 B. H_2O and O_2 C. SO_2 and H_2 D. SO_2 and O_2

C 13. When electric current is passed through acidified water for 1930 sec , 1120 mL of hydrogen gas is collected at STP at the cathode , what is the current passed in amperes ?

A. 0.05 B. 0.5 C. 5 D. 50

A 14. A galvanic cell with reduction potential of A= 2.23 v, and B = -1.43 v . The value of E^{o}_{Cell} is

A. 3.66 v B. 0.8 v C. -0.8 v D. -3.66 v

C 15. Reduction potential of A,B, C and D are 0.8v, 0.79v, 0.34v, and -2.34 v respectively. Which element displaces all the other three elements?

A. B B. A C. D D. C

B 16. When an electrolytic solution conducts electricity, current is carried out by

A. Electron B. cations and anions C. neutral atoms D. none

A 17. E^{o}_{red} for $F_2 + 2e^- \longrightarrow 2F^-$ is 2.8 v, E^{o}_{red} for $\frac{1}{2}F_2 + e^- \longrightarrow F^-$ is

A. 2.8v B. 1.4 v C. -2.8 v D. -1.4 v

A 18. The oxidation number of S in $Na_2S_4O_6$ is

A. 5/2 B. 3 C. 5 D. 2

A 19. In a cell containing Zn electrode and standard hydrogen electrode (SHE) , the zinc electrode acts as

A. Anode B. Cathode C. Neither cathode nor anode D. both anode and cathode

A 20. 1 mole of Al is deposited by X coulombs of electricity passing through $Al(NO_3)_3$ solution . The number of moles of silver deposited by X coulombs of electricity from $AgNO_3$ solution is

A. 3 B. 4 C. 2 D. 1

A 21. When electroplating a copper medal with silver, the mass of the negative electrode

A. Increases B. Decreases C. remains the same D. can't be determined

D 22. What is the role of the salt bridge

A. To increase the salt voltage

B. To connect the two electrodes

C. To provide a source of reactants

D. To allow ion flow and maintain electrical neutrality

D 23. Which of the following is true about the electrolysis of a molten MX ?

A. X is produced at the negative electrode

B. M is produced at the anode

C. Electrons are not transferred from one species to the other

D. The overall reaction is redox

B 24. By passing 9.65 A current for 16 min and 40 sec , the volume of oxygen gas liberated at STP will be

A. 280 mL B. 560 mL C. 1120mL D. 2240mL

B 25. The mathematical expressions for faraday's first law of electrolysis is summarized as:-

A. $\frac{mIt}{MF} = n$ B. $m = \frac{EQ}{F}$ C. $M = \frac{mIt}{nF}$ D. $M = It/F$